

Procedural Knowledge Generation with Selective Retrieval from Past Outputs

Anonymous submission

Abstract

Retrieval-augmented generation (RAG) has become a common approach to ground the output of large language models (LLMs), especially when the model has not been trained on the topic at hand. RAG requires careful search tuning to increase the chance that the model sees the documents necessary to answer the question, especially if the necessary information is dispersed across many documents. For topics unseen in training, many of the same documents will be needed for all queries in order to recreate basic knowledge. If we instead answer queries one at a time and store the outputs in a library, much of the information needed for later outputs will be densely available in the earlier responses. We propose a new augmentation technique called Selective Retrieval from Past Outputs (SRPO) which retrieves relevant texts from the model’s previous outputs, and filters by asking the model if the text is useful for this particular generation.

To demonstrate that this approach more easily surfaces the necessary information, we publish a new dataset called LC-Step containing 120 clean, step-by-step procedures extracted from the LangChain Python documentation (created after the OpenAI knowledge cutoff date), along with a model-based evaluation metric that judges the quality of generated procedures. Our study shows that GPT-4 cannot reliably generate these procedures when given a candidate goal, even when augmented with retrieved texts from LangChain’s conceptual documentation and API reference. We demonstrate that collecting outputs into a library of retrievable procedures can provide further benefit to later generations, as long as procedures retrieved from the library are highly relevant to the current generation.

References